

Per-stage runtime of the four-index BIE pipeline on the graphene-monolayer system (precompute-once volume field)

Numerical experiment 6.5 — exp65_orbital_bench

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1 Setup

Single right-hand-side solve of the screened-Coulomb four-index integral on the real graphene monolayer (Wannier p_z orbital, k_{323201}), using the production pipeline with the `PrecomputedVolumeField` (`cache_fft`) volume-field method for the volume-potential stages. All timings are warm (post-JIT), measured on a single idle node.

Hardware / software	
CPU	2× AMD EPYC 9474F (96 cores), 96 threads
Memory	1.50 TB (peak RSS 29.40 GB)
OS / kernel	Rocky Linux 8.10, kernel 6.6
Julia	1.12.6
Libraries	FINUFFT.jl 3.5.2, finufft_jll 2.5.1, FMM3D
Problem	
Geometry	slab $L \times L \times L_z = 90 \times 90 \times 3.35$, $\varepsilon_{\text{in}} = 3.5$
Source	squared p_z orbital, 355 862.00 grid points
Tolerances	source/RHS 10^{-3} , LHS/eval 10^{-5} , GMRES 10^{-5}
Discretization	$n_{\text{quad}} = 6$, edge level 4, edge correction on
Interface	960 768.00 points; GMRES 8 iters, residual 7.10×10^{-6}
Result	$u_{\text{onsite}} = 9.88 \text{ eV}$

2 Per-stage runtime

3 Notes

- **Dominant costs** are GMRES (22.20 s), LHS assembly (13.60 s), field construction (8.80 s), and the interface build (6.80 s); the pipeline is solve/assembly-bound.
- **Amortization.** The RHS-independent precompute (field construction + interface build + LHS assembly = 29.10 s) is paid once. Each additional right-hand side on the same geometry/source family costs RHS + GMRES + eval ≈ 30.00 s, dominated by GMRES.
- **Volume field.** The precomputed spectral field is reused across all mesh-refinement depths and the evaluation; with `cache_fft` the u_{int} read-out is interpolation-only (0.12 s).

Phase	Stage	Time (s)
Load	XSF read + truncation + screening	3.25
Precompute	Field construction (<code>cache_fft</code>)	8.82
Precompute	Interface build (RHS-adaptive)	6.76
Precompute	LHS assembly (near corrections)	13.56
Solve	RHS assembly (field)	2.18
Solve	GMRES solve (8 iterations)	22.21
Eval	Volume potential u_{int} (field)	0.12
Eval	Scattered potential (FMM + cubature)	5.38
Total		62.28

Table 1: Wall-clock time per pipeline stage (warm, 96 threads).

Phase	Time (s)	Share (%)
Load	3.25	5.20
Precompute (construct + build + LHS)	29.14	46.80
Solve (RHS + GMRES)	24.39	39.20
Eval (u_{int} + scatter)	5.50	8.80
Total	62.28	100.00

Table 2: Phase totals.

- **Memory.** Peak RSS 29.40 GB, set by the `cache_fft` fine-grid storage; the standard field (`cache_fft = false`) peaks at ~ 14.00 GB with $u_{\text{int}} \approx 0.40$ s.

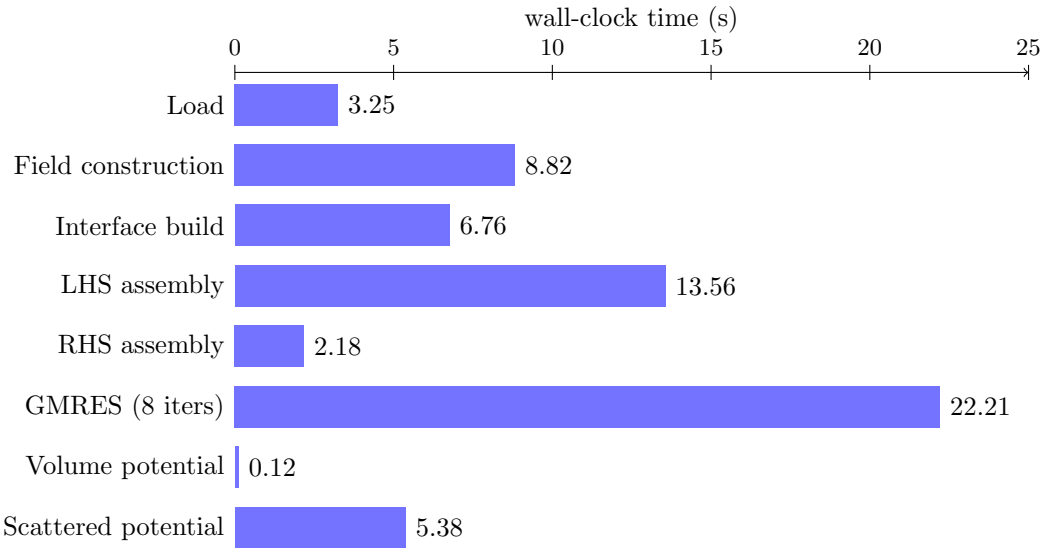


Figure 1: Per-stage wall-clock time. The pipeline is dominated by GMRES, LHS assembly, and the one-time field construction; the volume-potential evaluation (u_{int}) is essentially free.